Home Lab Project Week 9, Day 3 Revised

Task:

Create and configure a third OPNsense interface for a DMZ network, connect it to its own VirtualBox network with its own IP range, and move the Ubuntu VM into it. No traffic rules between hosts are written today; those come in Week 10.

Why a DMZ is needed

A firewall can only filter traffic that passes through it. While Kali and Ubuntu shared the same LAN subnet, they talked directly through the virtual switch, and no OPNsense rule could affect that traffic.

A DMZ (demilitarized zone) is a separate network for servers. Putting Ubuntu there forces every packet between Kali and Ubuntu through OPNsense, so later rules can control it. If the vulnerable Ubuntu server is compromised, the firewall still stands between it and the management LAN.

Starting point (before)

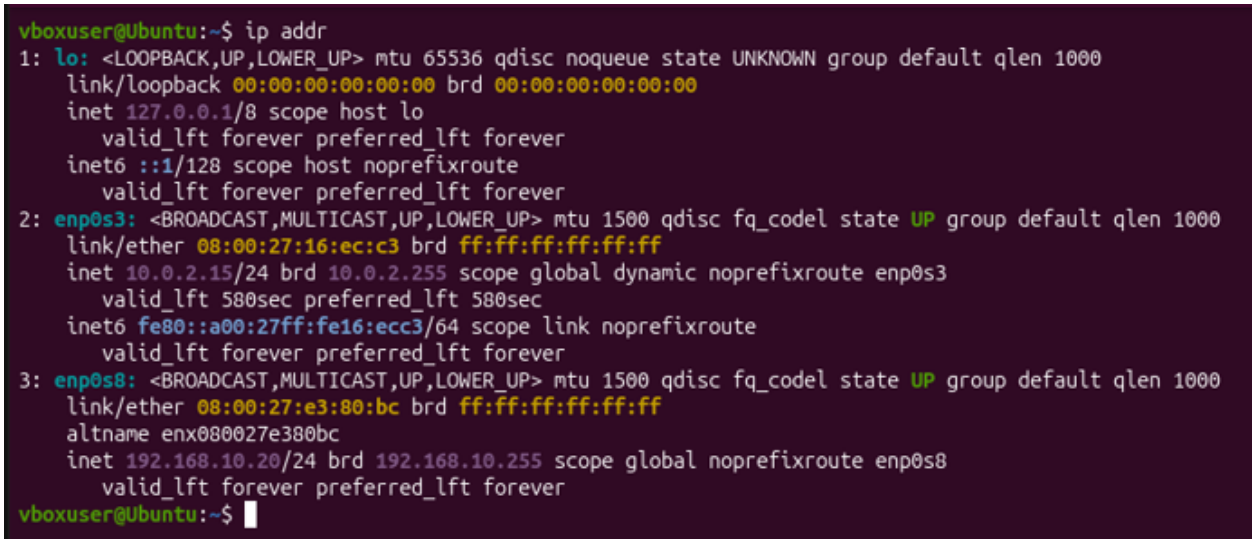
Both VMs sat on the same LAN, built in earlier lab work:

Device	Interface / network	IP address
OPNsense WAN	VirtualBox NAT Network	DHCP (10.0.2.x)
OPNsense LAN	Internal Network `intnet`	192.168.10.1/24
Kali	LAN	192.168.10.10

Ubuntu	LAN	192.168.10.20

Observation: Kali and Ubuntu could ping each other even with the default allow LAN rules disabled, because same-subnet traffic never reaches OPNsense.

- Screenshot of `ip a` on Ubuntu showing 192.168.10.20 (before)



Target design (after)

OPNsense interface	VirtualBox adapter	Attached to	Firewall IP	Host
WAN	Adapter 1	NAT Network	DHCP	Internet
LAN	Adapter 2	Internal Network `intnet`	192.168.10.1/24	Kali 192.168.10.10
DMZ (new, OPT1)	Adapter 3	Internal Network `Ubuntu-Net`	192.168.20.1/24	Ubuntu 192.168.20.10

[embed: node/fcf37f7f-847a]

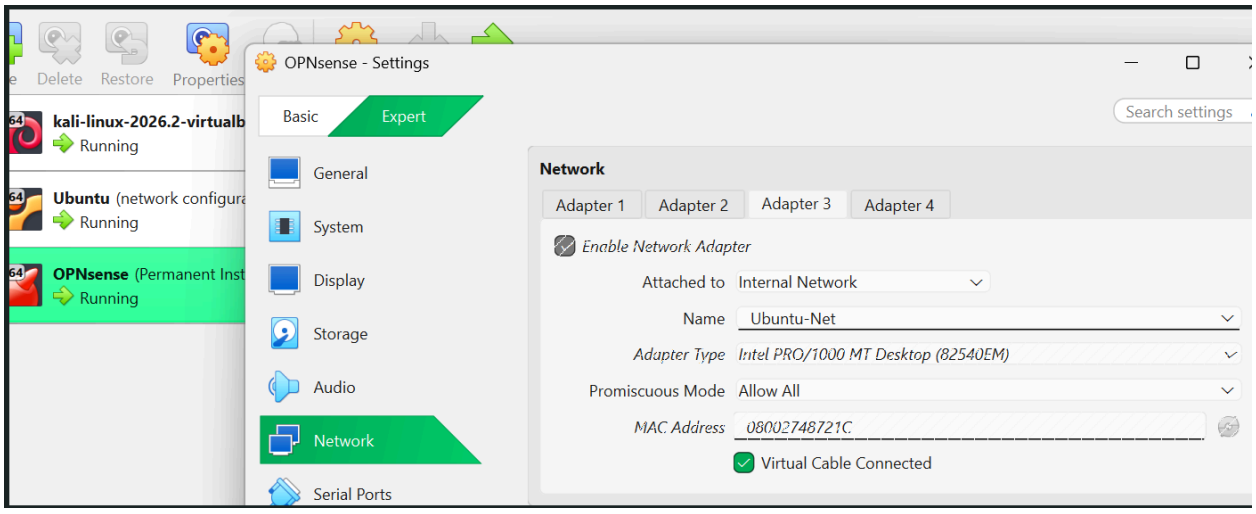
Kali stays on the LAN. Ubuntu moves to the new DMZ, so traffic between them must cross OPNsense.

Step 1: VirtualBox adapter for the DMZ

1. Shut down the OPNsense VM. Adapter settings cannot change while it runs.
2. Open Settings > Network > Adapter 3 and set:
    - Enable Network Adapter: checked
    - Attached to: Internal Network
    - Name: `Ubuntu-Net`
    - Advanced > Cable Connected: checked
3. Write down the adapter's MAC address (Advanced section). It identifies the interface in OPNsense.
4. Click OK and start OPNsense.

The internal network name must match exactly on OPNsense and Ubuntu, including capital letters. A different spelling creates a second, separate network.

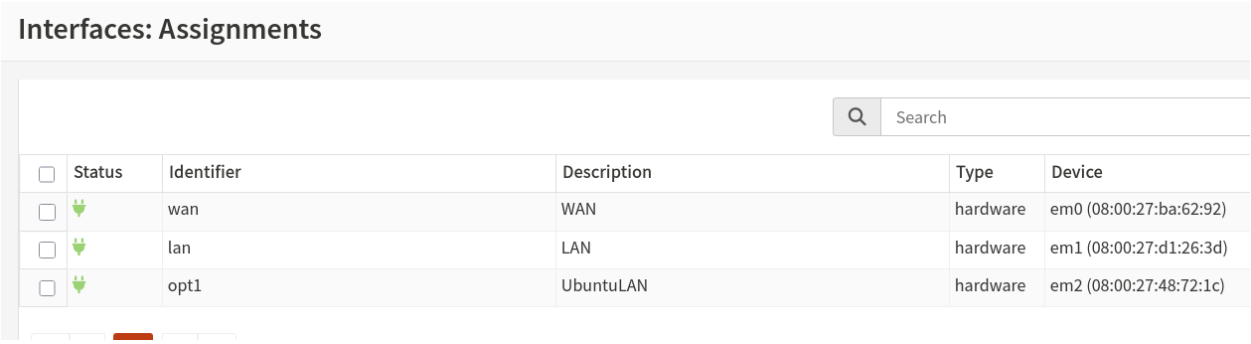
- Screenshot of Adapter 3 settings



Step 2: Assign and configure the DMZ interface

This is done in the OPNsense web GUI from Kali (https://192.168.10.1).

1. Go to Interfaces > Assignments.
2. Select the new device, `em2`, and confirm it by the MAC address **08:00:27:48:72:1c** from Step 1.



3. We select OPT1 by clicking on the pencil in front of it to edit and pressing the orange plus sign. Then we click save it.
4. Open Interfaces > [OPT1] and set:

Field	Value	Why
Enable	Checked	Turns the interface on
Description	`DMZ`	Menus, rules and logs show OPT1 instead of DMZ
Block private networks	Unchecked	WAN-only setting; 192.168.x is private
Block bogon networks	Unchecked	WAN-only setting
IPv4 Configuration Type	Static IPv4	Firewall interfaces need fixed addresses
IPv6 Configuration Type	None	Not used in this lab

IPv4 address	`192.168.20.1` / `24`	The DMZ gateway; .1 is the firewall, not a host
IPv4 Upstream Gateway	None	Only WAN has an upstream gateway

5. Click Save, then Apply changes.

- Screenshot of Interfaces > Assignments showing WAN, LAN and DMZ
- Screenshot of the DMZ interface settings

Interfaces: [DMZ]

Basic configuration

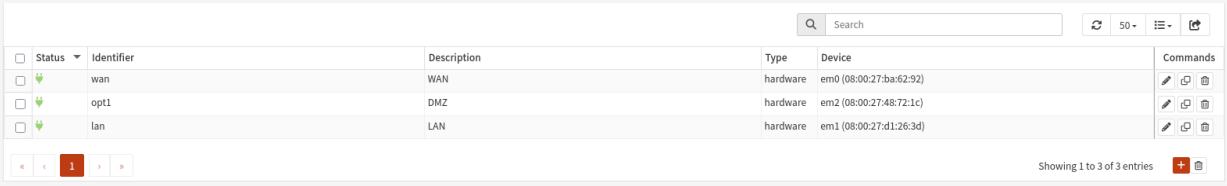
Enable	☑ Enable Interface
Lock	☑ Prevent interface removal
Identifier	opt1
Device	em2
Description	DMZ

Generic configuration

Block private networks	☐
Block bogon networks	☐
IPv4 Configuration Type	Static IPv4
IPv6 Configuration Type	None
MAC address	
Promiscuous mode	☐
MTU	

OPNsense (c) 2014-2026 Deciso B.V.

Interfaces: Assignments

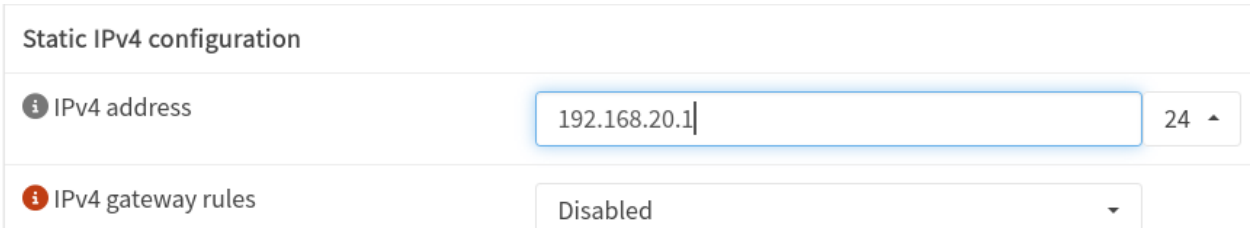


Search 50

Status	Identifier	Description	Type	Device	Commands
	wan	WAN	hardware	em0 (08:00:27:ba:62:92)	
	opt1	DMZ	hardware	em2 (08:00:27:48:72:1c)	
	lan	LAN	hardware	em1 (08:00:27:d1:26:3d)	

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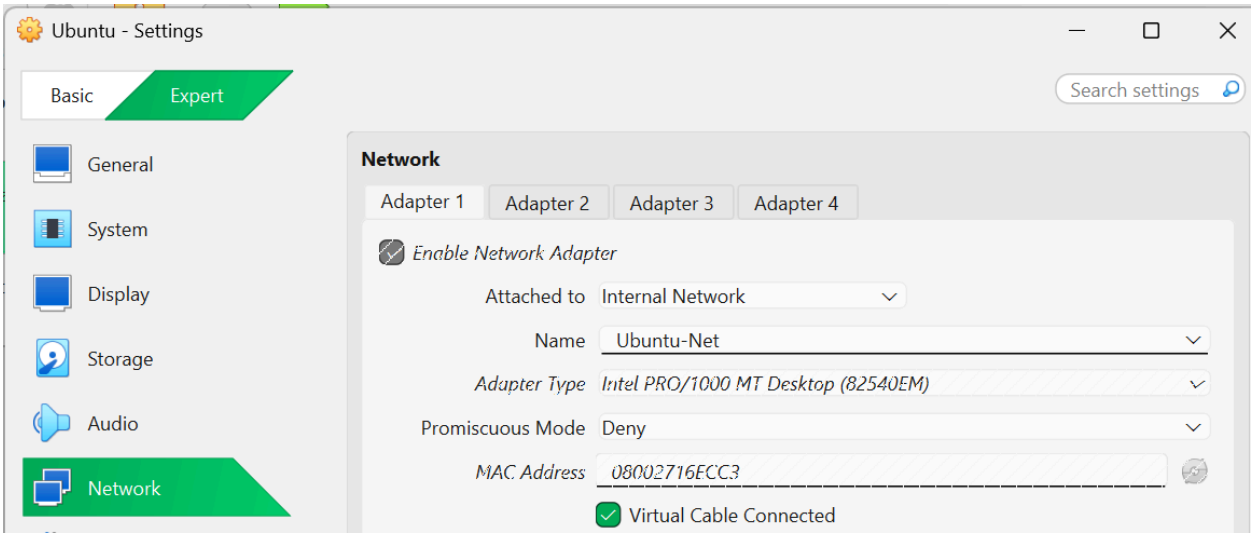
Showing 1 to 3 of 3 entries



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Step 3: Move Ubuntu into the DMZ

1. Shut down Ubuntu. In Settings > Network, attach its LAN adapter to Internal Network `Ubuntu-Net`, then start Ubuntu.
2. Find the adapter and its connection name. Match the MAC address from `ip link` to the adapter on `Ubuntu-Net`:





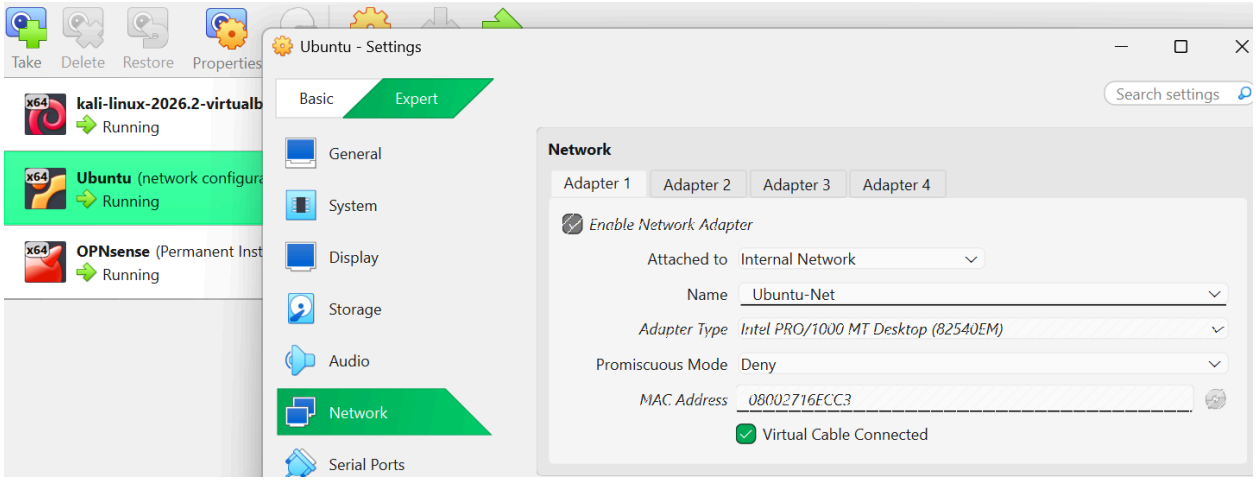
3. Set the Static IP. Ubuntu.

sudo nmcli con mod "<connection name>" ipv4.method manual ipv4.addresses 192.168.20.10/24 +ipv4.gateway 192.168.20.1"

sudo nmcli con up "<connection name>"

Because the DMZ adapter is Ubuntu's only adapter, let's set `ipv4.gateway 192.168.20.1` instead of the route. NetworkManager saves the settings, so they survive a reboot.

- Screenshot of Ubuntu's VirtualBox network settings



Step 4: Starter Rule for the DMZ

A new interface may block all traffic by default, including ping to its own gateway.

We will try to ping

The ping succeeded.

If the ping was blocked, we need to go on Firewall , choose Rules, interface DMZ, and change select pass for action, ICMP for protocol, UbuntuLan for source.

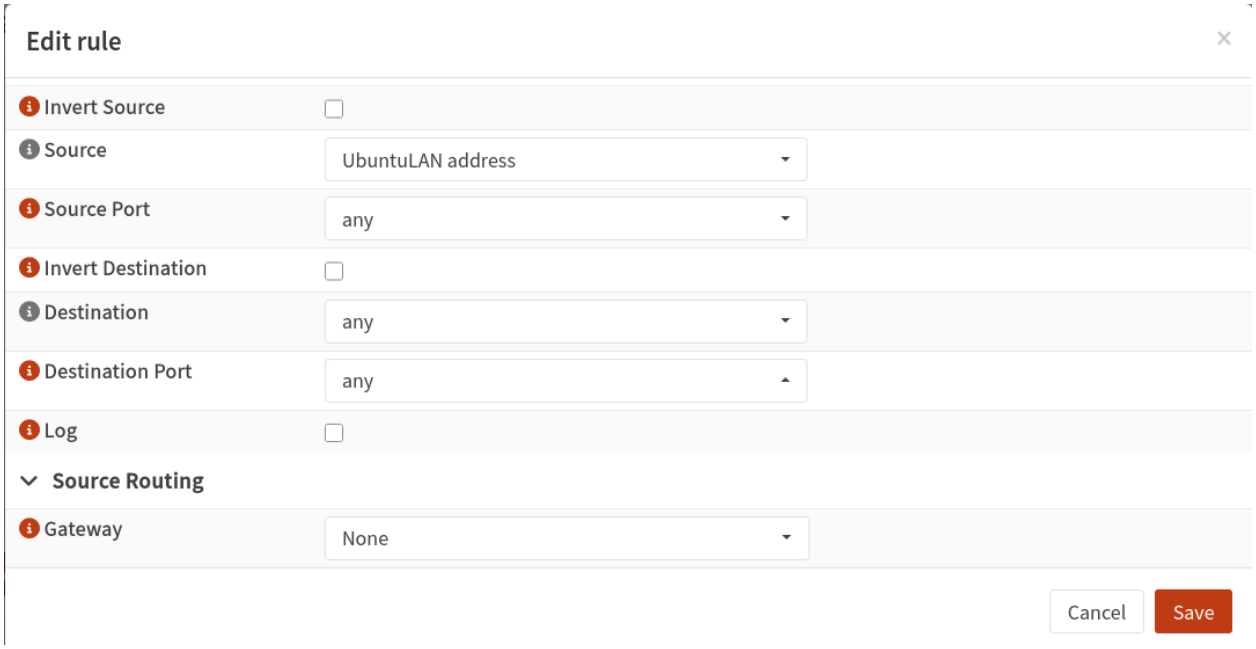
```
vboxuser@Ubuntu:~$ ping -c 4 192.168.20.1
PING 192.168.20.1 (192.168.20.1) 56(84) bytes of data.
64 bytes from 192.168.20.1: icmp_seq=1 ttl=64 time=6.90 ms
64 bytes from 192.168.20.1: icmp_seq=2 ttl=64 time=3.19 ms
64 bytes from 192.168.20.1: icmp_seq=3 ttl=64 time=3.56 ms
64 bytes from 192.168.20.1: icmp_seq=4 ttl=64 time=1.97 ms

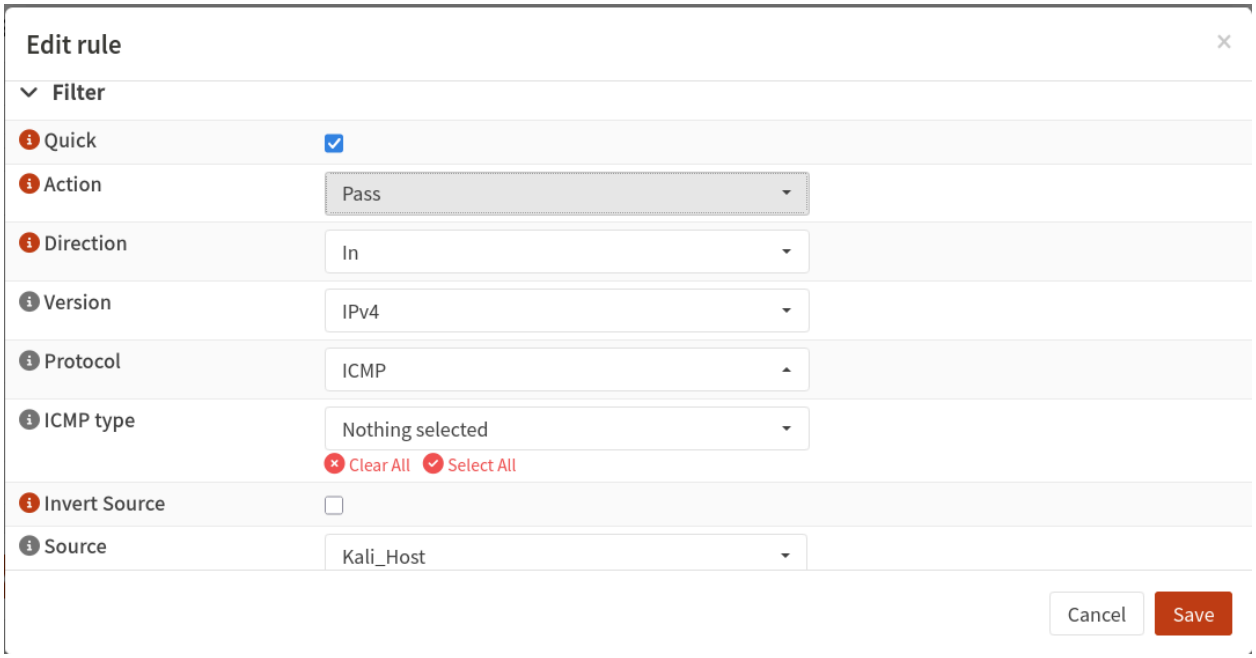
--- 192.168.20.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3008ms
rtt min/avg/max/mdev = 1.973/3.904/6.896/1.824 ms
vboxuser@Ubuntu:~$
```

Action	Protocol	Source	Destination	Description
Pass	ICMP (Echo request)	DMZ net	DMZ address	Allow DMZ hosts to ping their gateway

Click Save, then Apply. Rules controlling traffic between Kali and Ubuntu are created in Week 10.

- Screenshot of the DMZ rule





Verification

These checks confirm the DMZ is built and connected. Reachability between Kali and Ubuntu is tested.

ID	Check	Where / command	Expected	Actual	Result
V1	DMZ interface up	Interfaces > Overview	DMZ up, 192.168.20.1/24		
V2	Ubuntu address	Ubuntu: `ip a`	192.168.20.10/24 on the DMZ adapter		
V3	Route to the LAN	Ubuntu: `ip route get 192.168.10.10`	via 192.168.20.1		

V4	Ubuntu reaches its gateway	Ubuntu: `ping -c 3 192.168.20.1`	3 replies with 0 packets loss
V5	Firewall reaches Ubuntu	Interfaces > Diagnostics > Ping, host 192.168.20.10	Replies with 0 packets loss
V6	Network names match	VirtualBox settings for OPNsense Adapter 3 and Ubuntu	Both `Ubuntu-Net`



```
 0) Logout                              7) Ping host
 1) Assign interfaces                   8) Shell
 2) Set interface IP address            9) pfTop
 3) Reset the root password            10) Firewall log
 4) Reset to factory defaults          11) Reload all services
 5) Power off system                   12) Update from console
 6) Reboot system                      13) Restore a backup

Enter an option: 7


Enter a host name or IP address: 192.168.20.10

PING 192.168.20.10 (192.168.20.10): 56 data bytes
64 bytes from 192.168.20.10: icmp_seq=0 ttl=64 time=5.059 ms
64 bytes from 192.168.20.10: icmp_seq=1 ttl=64 time=2.810 ms
64 bytes from 192.168.20.10: icmp_seq=2 ttl=64 time=4.104 ms

--- 192.168.20.10 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 2.810/3.991/5.059/0.922 ms

Press ENTER to continue.
```

```
vboxuser@Ubuntu:~$ ping -c 4 192.168.20.1
PING 192.168.20.1 (192.168.20.1) 56(84) bytes of data.
64 bytes from 192.168.20.1: icmp_seq=1 ttl=64 time=6.90 ms
64 bytes from 192.168.20.1: icmp_seq=2 ttl=64 time=3.19 ms
64 bytes from 192.168.20.1: icmp_seq=3 ttl=64 time=3.56 ms
64 bytes from 192.168.20.1: icmp_seq=4 ttl=64 time=1.97 ms

--- 192.168.20.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3008ms
rtt min/avg/max/mdev = 1.973/3.904/6.896/1.824 ms
```

```
vboxuser@Ubuntu:~$ ip route get 192.168.10.10
192.168.10.10 via 192.168.20.1 dev enp0s3 src 192.168.20.10 uid 1000
    cache
```

Evidence checklist

- Ubuntu `ip a` before the move (192.168.10.20)
- VirtualBox Adapter 3 on OPNsense (`Ubuntu-Net`)
- Interfaces > Assignments with WAN, LAN and DMZ

- DMZ interface settings page
- Interfaces > Overview with DMZ up
- DMZ starter rule
- Ubuntu `ip a`, `ip route get` and ping to 192.168.20.1 after the move